

Supplementary Material (Appendix)

Remote Sensing Image Change Caption Generation for Indian Sentinel-2 Imagery

Sankalp Chaturvedi (2025201083), KSPSVLN Siddardha Kumar Kavuri (2025201061), Rinkesh Verma (2025201070) · Mentor: Rama Chandra Prasad

Appendix tables and figures are cited in the main text as Appendix Table A_n and Appendix Figure A_n .

Prediction sheets for all saved test samples are published at

kaggle.com/datasets/kpsvlnsiddardha/rsicc-model-run-reports.

Appendix tables

Appendix Table A1. Full LEVIR-CC test metrics ($n = 1,929$).

Model	Loss	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	Cosine
Phase 1	0.9690	0.4720	0.3855	0.3393	0.3027	0.5966	0.6214	0.6477
Phase 2	0.9345	0.5168	0.4237	0.3712	0.3284	0.6304	0.6545	0.7061
Final	0.8982	0.5311	0.4355	0.3810	0.3365	0.6392	0.6604	0.7123
Phase 6	0.9225	0.5214	0.4297	0.3777	0.3358	0.6357	0.6624	0.7149
Phase 8, stage 1a (epoch 7)	—	0.5092	0.4167	0.3658	0.3250	0.6155	0.6482	0.7007
Phase 8, stage 1b	—	0	0	0	0	0	0	0.0104

Appendix Table A2. Training configuration.

Item	Phase 1	Phase 2	Final	Phase 6	Phase 8 (1a)	Phase 8 (1b)
Input size (px)	256	224	224	4 × 224 tiles	224	224
Batch size	16	16	16	16	8	8
Epochs	≥ 10	10	15	15	15	10
Learning rate	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}	5×10^{-5}
Weight decay	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
Vocabulary	733	733	1,300	1,300	733	Qwen tokenizer

Appendix Table A3. Forgetting after SECOND-CC fine-tuning (LEVIR-CC test set).

Model	Loss	BLEU-4	METEOR	ROUGE-L	Cosine
Final, before	0.898	0.337	0.639	0.660	0.712
Final, after	4.080	0.149	0.394	0.437	0.458
Phase 6, before	0.923	0.336	0.636	0.662	0.715
Phase 6, after	3.520	0.121	0.318	0.374	0.438

Appendix Table A4. Prediction behaviour on the saved prediction sheets (300 samples each).

Prediction sheet	Changed pairs called “no difference”	Unchanged pairs correct	Distinct captions
Phase 6, LEVIR-CC	43 / 164	98 / 136	21
Phase 8, LEVIR-CC	31 / 164	96 / 136	32
Final after fine-tuning, LEVIR-CC	164 / 164	136 / 136	1
Phase 6, SECOND-CC	16 / 255	33 / 45	97
Phase 8, SECOND-CC	19 / 255	30 / 45	82

Appendix Table A5. Accepted and rejected patch pairs by region.

Region	Accepted	Rejected
Deccan	3,902	2,738
Central-east	3,179	2,485
North plains	2,655	2,129
West desert	2,095	2,033
North-east	1,580	804
Himalaya	1,529	1,095
South and islands	670	994
Total	15,610	12,278

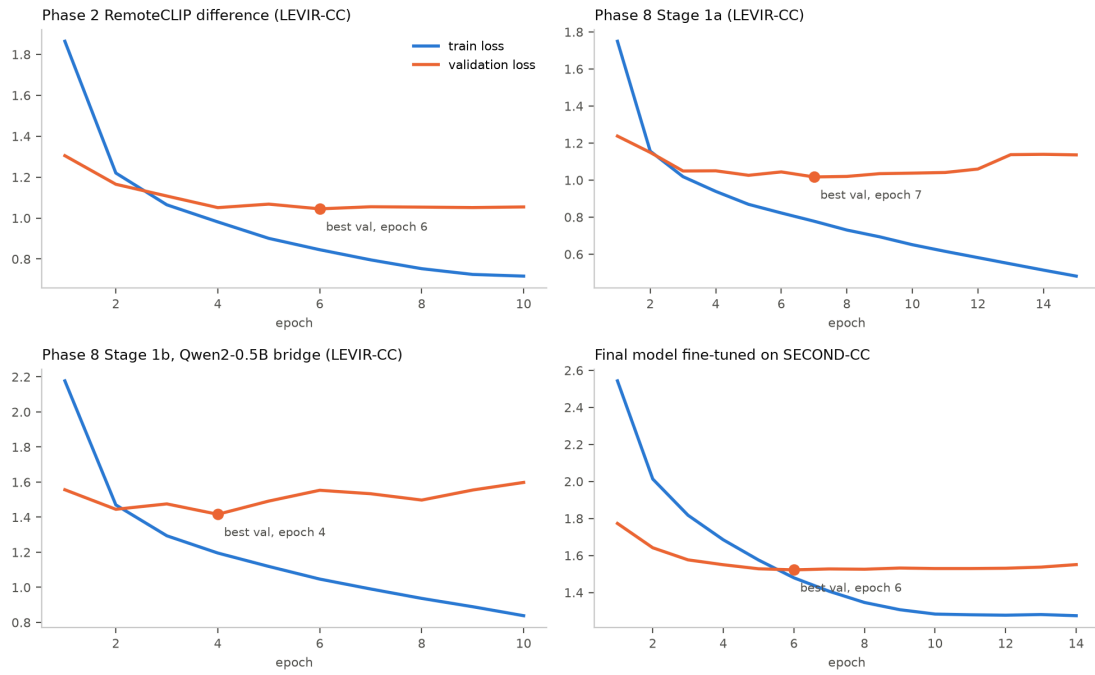
Appendix Table A6. How often each model called a pair “change”, by human label (n = 495).

Human label	n	Qwen3-VL-32B	Qwen3-VL-235B
Seasonal	214	200 (93%)	67 (31%)
No change	226	184 (81%)	53 (23%)
Change	55	49 (89%)	25 (45%)

Appendix Table A7. Results on all labelled pairs (n = 962; human label where available, automatic label otherwise).

Rule	Recall	Precision	False-change rate
Qwen3-VL-32B	96%	26%	47%
Qwen3-VL-235B	79%	48%	15%
Both agree	78%	51%	13%

Appendix figures



Appendix Figure A1. Training and validation loss for four runs: Phase 2, Phase 8 stage 1a, Phase 8 stage 1b and the Final model fine-tuned on SECOND-CC. Dots mark the best validation epoch.



LEVIR-CC test sample 4

Reference:
the vegetation has been replaced by a road and many villas around .
Phase 6 run:
a huge residential area is built in almost whole scene
Phase 8 run:
the vegetation quantity is increased in many parts of the scene
Final, after SECOND-CC fine-tuning:
there is no difference



LEVIR-CC test sample 17

Reference:
a road appears at the bottom and many houses are scattered replacing the trees .
Phase 6 run:
a huge residential area is built in almost whole scene
Phase 8 run:
many buildings are constructed in the middle of the scene
Final, after SECOND-CC fine-tuning:
there is no difference



LEVIR-CC test sample 19

Reference:
the trees have been replaced by many buildings .
Phase 6 run:
a huge residential area is built in almost whole scene
Phase 8 run:
many buildings are constructed in the middle of the scene
Final, after SECOND-CC fine-tuning:
there is no difference



LEVIR-CC test sample 42

Reference:
a house appears at the top right corner .
Phase 6 run:
there is no difference
Phase 8 run:
there is no difference
Final, after SECOND-CC fine-tuning:
there is no difference



LEVIR-CC test sample 2

Reference:
there is no difference .
Phase 6 run:
the vegetation quantity is decreased in many parts of the scene
Phase 8 run:
there is no difference
Final, after SECOND-CC fine-tuning:
there is no difference

Appendix Figure A2. The same LEVIR-CC test pairs captioned by three runs, with the reference caption.

SECOND-CC test samples (Phase 6 run)



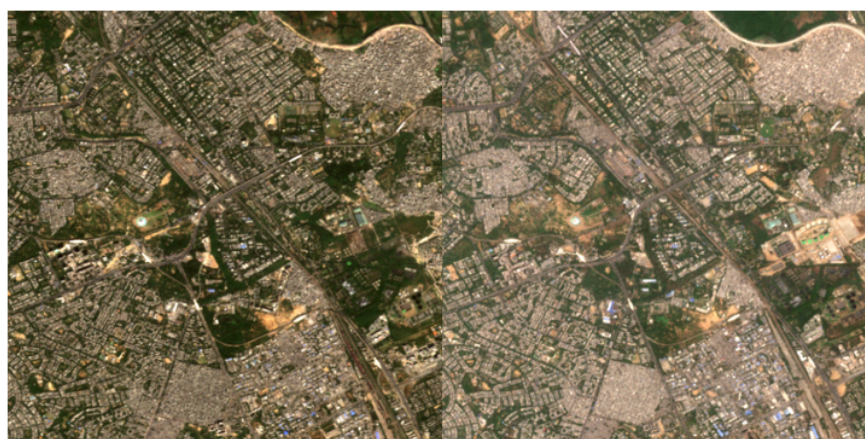
Appendix Figure A3. SECOND-CC test predictions (Phase 6 run).



Appendix Figure A4. Whole-image (left) and patch-wise (right) inference on unlabelled high-resolution pairs.



Appendix Figure A5. A pair from the pilot Indian Sentinel-2 collection with its absolute-difference map.



Before

After

Owen2.5-VL: "...increased urban density and expansion of buildings, particularly along the central axis, indicating rapid development. There's also visible growth in road infrastructure..."

TeoChat: "Nothing meaningful changed."

Appendix Figure A6. An Indian pilot pair with the answers of two off-the-shelf captioners.

Plain prompt: change found (correct). Human label: Change (bihar_004_r2c1)



Before

After

Pixel boxes: answer 'none' (missed)



Before, pixel-difference boxes

After, pixel-difference boxes

CNN boxes: answer 'seasonal' (missed)

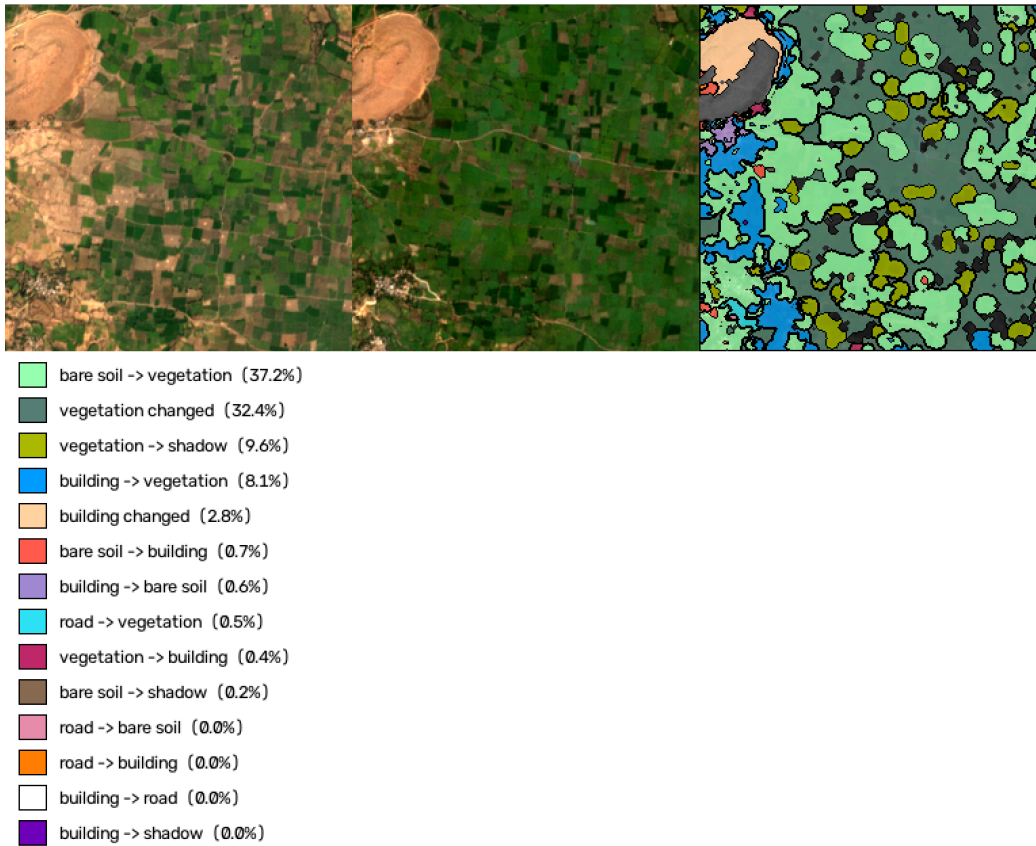


Before, CNN boxes

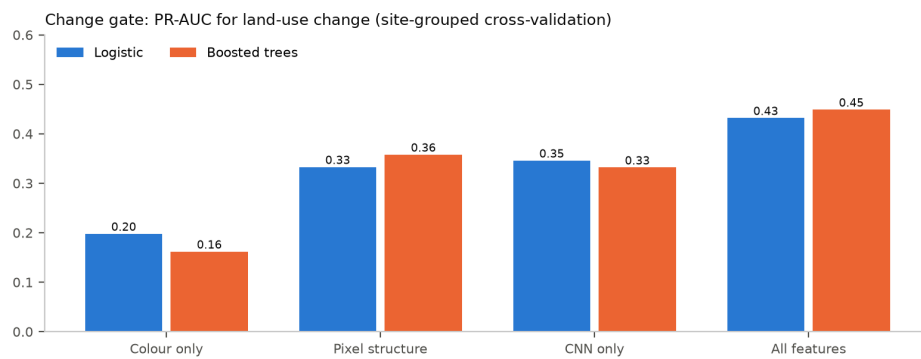
After, CNN boxes

Appendix Figure A7. A human-confirmed change with pixel-difference and CNN region boxes; the plain question finds the change, the boxed prompts miss it.

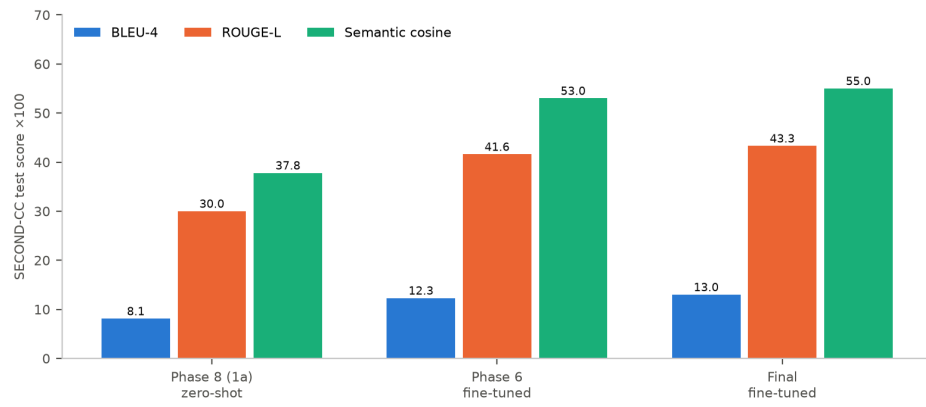
Human label: Seasonal (madhya_pradesh_071_r3c3). Before | After | change map
 328 caption with map: "Large areas of bare soil became vegetated, vegetation changed across the scene, and buildings were removed in the left and bottom-left, leaving vegetation."



Appendix Figure A8. Rule-based change map on a pair labelled seasonal, with the caption written from it.



Appendix Figure A9. Change gate: precision–recall area under the curve (PR-AUC) per feature set, site-grouped cross-validation.



Appendix Figure A10. SECOND-CC test scores: zero-shot (Phase 8 stage 1a) and after fine-tuning (Phase 6, Final).